

*MA.6.NSO.2.1***Benchmark**

MA.6.NSO.2.1 **Multiply and divide positive multi-digit numbers with decimals to the thousandths, including using a standard algorithm with procedural fluency.**

Benchmark Clarifications:

Clarification 1: Multi-digit decimals are limited to no more than 5 total digits.

Connecting Benchmarks/Horizontal Alignment

- MA.6.NSO.3.3
- MA.6.AR.1.1, MA.6.AR.1.3
- MA.6.AR.2.4
- MA.6.AR.3
- MA.6.GR.2
- MA.6.DP.1.2, MA.6.DP.1.3
- MA.7.NSO.2.2

Terms from the K-12 Glossary

- Area Model
- Commutative Property of Multiplication
- Expression
- Dividend
- Divisor

Vertical Alignment**Previous Benchmarks**

- MA.5.NSO.2.1, MA.5.NSO.2.2, MA.5.NSO.2.4, MA.5.NSO.2.5

Next Benchmarks

- MA.8.NSO.1.5, MA.8.NSO.1.7

Purpose and Instructional Strategies

In previous courses, students multiplied and divided multi-digit whole numbers and represented remainders as fractions. They also estimated and determined the product and quotient of multi-digit numbers with decimals to the hundredths and multiplied and divided the product and quotient of multi-digit numbers with decimals to the hundredths by one-tenth and one-hundredth with procedural reliability. In grade 6 accelerated, students multiply and divide positive rational numbers with procedural fluency, including dividing numerators by denominators to rewrite fractions as decimals. Students will also become fluent in all operations with positive and negative rational numbers. In future courses, students will use the order of operations with rational numbers, including exponents and radicals to solve multi-step mathematical and real-world problems.

- Instruction includes representing multiplication in various ways.
 - $3.102 \times 1.1 = 3.4122$
 - $(3.102)(1.1) = 3.4122$
 - $3.102(1.1) = 3.4122$
 - $3.102 \cdot 1.1 = 3.4122$

- Students should continue demonstrating their understanding from previous courses that division can be represented as a fraction.
- A standard algorithm is a systematic method that students can use accurately, reliably and efficiently (no matter how many digits) depending on the content of the problem. It is not the intention to require students to use a standard algorithm all of the time. However, students are expected to become fluent with a standard algorithm by certain grade levels as stated within the benchmarks.
- Instruction includes a variety of methods and strategies to multiply and divide multi-digit numbers with decimals.
 - Area Models

$$3.102 \times 1.1 = 3.4122$$

	3	0.1	0.00	0.002
1	3	0.1	0.00	0.002
0.1	0.3	0.01	0.000	0.0002

- Partial Products

$$\begin{array}{r}
 3.102 \\
 \times 1.1 \\
 \hline
 0.3102 \\
 + 3.1020 \\
 \hline
 3.4122
 \end{array}$$

- Multiplying as if the factors are whole numbers and applying the decimal places to the final product based on the number of decimals represented in the factors (*MTR.3.1*).

$$\begin{array}{ccc}
 \begin{array}{r} 3.102 \\ \times 1.1 \\ \hline \end{array} & \longrightarrow & \begin{array}{r} 3102 \\ \times 11 \\ \hline 3102 \\ + 31020 \\ \hline 34122 \end{array} & \longrightarrow & \begin{array}{l} \text{4 decimal places in factors} \\ 3.4122 \end{array}
 \end{array}$$

- Students should develop fluency with and without the use of a calculator when performing operations with positive decimals.

Common Misconceptions or Errors

- Students may incorrectly apply rules for adding or subtracting decimals to multiplication of decimals, believing place values must be aligned.
- Students may confuse the lining up of place values when multiplying or dividing vertically by omitting or forgetting to include zeros as placeholders in the partial products or quotients.

Strategies to Support Tiered Instruction

- Instruction includes the use of estimation to ensure the proper placement of the decimal point in the final product or quotient of decimals.
 - For example, if finding the product of 12.3 and 4.8, students should estimate the product to be close to 60, by using 12 and 5 as friendly numbers, then apply the decimal to the actual product of 123 and 48, which is 5904. Based on the estimate, the decimal should be placed after 59 to produce 59.04.
- Teacher encourages students with a firm understanding of multiplying and dividing fractions to convert decimal values into equivalent fraction forms and then perform the desired operation and convert the solution back to decimal form.
- Teacher provides graph paper to utilize while applying an algorithm for multiplying or dividing to keep numbers lined up and help students focus on place value.
- Instruction includes providing opportunities to reinforce place values with the use of base ten blocks or hundredths grids.

Instructional Tasks

Instructional Task 1 (MTR.6.1)

Carlos spent \$20.76 on chips when his friends came over. Each bag of chips costs \$3.46 and each bag has 3 servings. What is the maximum number of friends that Carlos can have over if each person can have a single serving of chips?

Instructional Task 2 (MTR.7.1)

Samantha has 6.75 bags of candy. A full bag of candy contains 13.125 ounces of candy. How many ounces of candy does Samantha have?

Instructional Task 3 (MTR.4.1, MTR.5.1)

Part A. Complete the table below using a calculator.

Expression	Solution	Expression	Solution
559(5)		5.59(5)	
325(25)		3.25(2.5)	
19(93)		19(9.3)	

Part B. Talk with a partner about what you notice from the table in Part A.

Part C. How are the expressions without decimals related to the expressions with decimals?

Is there a relationship between the decimal placements in the expressions and the solutions?

Part D. If $2368(421) = 996,928$, what would you expect $2.368(4.21)$ be equal to?

Instructional Items

Instructional Item 1

Determine the product of 23.5 and 2.3.

Instructional Item 2

The expression $13.31 \div 0.125$ is equivalent to what number?

Instructional Item 3

Determine the quotient of 201.3 and 1.83.

Instructional Item 4

The expression 4.321×2.3 is equivalent to what number?

**The strategies, tasks and items included in the BIG-M are examples and should not be considered*